Subject Description Form

Subject Code	EIE3333
Subject Title	Data and Computer Communications
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To provide solid foundation to students about the architectures and operations of communication networks. 2. To enable students to master the knowledge about computer networking in the context of real-life applications. 3. To prepare students to learn and to critically evaluate new knowledge and emerging technology in communication networks.
Intended Subject Learning Outcomes	**Upon completion of the subject, students will be able to:** <u>Category A: Professional/academic knowledge and skills</u> 1. Demonstrate deep understanding of the services, functions, and inter-relationship of different layers in communication network models 2. Describe how components in different layers inter-operate and analyze their performance. 3. Explain and apply the principles and practices of communication networks. 4. Learn new techniques and to align new technologies to existing network infrastructure. <u>Category B: Attributes for all-roundedness</u> 5. Present ideas and findings effectively. 6. Learn independently.
Subject Synopsis/ Indicative Syllabus	**Syllabus:** 1. <u>Computer Networks, Services, and Layered Architectures</u> Evolution of networking and switching technology; protocol and services; layered network architectures: OSI 7-layer model and the TCP/IP architecture. 2. <u>Digital Transmission and Protocols in Data Link Layer</u> Line coding techniques, error detection and correction; automatic repeat request (ARQ) protocol and reliable data transfer service; sliding-window flow control; framing and point-to-point protocol; flow control and error controls; high level data link control (HDLC) protocol and point-to-point protocol (PPP). 3. <u>Local Area Networks (LANs) and Wireless LANs</u> Media Access Control (MAC) protocols: the IEEE802.3 Ethernet and IEEE802.11 wireless LAN standards; interconnection of LANs: bridge, switch, and virtual LAN. 4. <u>Network Layer Protocols</u> Network layer operations; connection oriented and connectionless services; Internet protocol (IP): IP datagram format, IP addressing, subnetting, IP routing

and router operations; Internet control message protocol (ICMP), dynamic host configuration protocol (DHCP); network address translation (NAT).

5. <u>Transport Layer Protocols</u>
   Transmission control protocol (TCP) and user datagram protocol (UDP).

Possible Laboratory Experiments:
1. Cisco router configuration and programming.
2. Static and Dynamic routing.
3. Network monitoring and analysis
4. Address resolution, ARP, IP, and TCP.

Teaching/ Learning Methodology

Teaching and Learning Method	**Intended Subject Learning Outcome**	**Remarks**
Lectures	1, 2, 3, 4	Fundamental principles and key concepts of the subject are delivered to students.
Tutorials	1, 2, 3, 4, 5	Supplementary to lectures. Students will be able to clarify concepts and to have a deeper understanding of the lecture material; Problems and application examples are given and discussed.
Laboratory sessions	3, 5, 6	Students will conduct practical exercises through network simulators to reinforce concepts and techniques learned.

Alignment of Assessment and Intended Subject Learning Outcomes

Specific Assessment Methods/ Task	**% Weighting**	**Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)**					
		1	**2**	**3**	**4**	**5**	**6**
1. Continuous Assessment	45%						
• Mid-Term Test	20%	✓	✓	✓	✓	✓	
• Assignment 1	10%	✓	✓	✓	✓	✓	
• Assignment 2	10%	✓	✓	✓	✓	✓	
• Laboratories	5%			✓		✓	✓
2. Examination	55%	✓	✓	✓	✓	✓	✓
Total	100%						

<table>
<tr><td></td><td colspan="2">Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

<table>
<tr><th>Specific Assessment Methods/ Tasks</th><th>Remark</th></tr>
<tr><td>Assignments, Tests and examination</td><td>These can measure the students' understanding of the theories and the concepts of the subject. End-of-chapter type problems used to evaluate students' ability in applying concepts and skills learnt in the classroom;
Assignments of reading report type to assess students' ability in acquiring new knowledge related to communication networks;
Students need to think critically and creatively in order to come up with a solution for an existing problem.</td></tr>
<tr><td>Laboratory sessions</td><td>Students are required to complete work sheets, to indicate their understanding and correct completion of the laboratories.</td></tr>
</table>
</td></tr>
<tr><td rowspan="7">Student Study Effort Expected</td><td colspan="2">Class contact (time-tabled):</td></tr>
<tr><td>• Lecture</td><td>24 Hours</td></tr>
<tr><td>• Tutorial/Laboratory/Practice Classes</td><td>15 hours</td></tr>
<tr><td colspan="2">Other student study effort:</td></tr>
<tr><td>• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination</td><td>36 Hours</td></tr>
<tr><td>• Tutorial/Laboratory/Practice classes: preview of materials, revision and/or reports writing</td><td>30 Hours</td></tr>
<tr><td>Total student study effort:</td><td>105 Hours</td></tr>
<tr><td>Reading List and References</td><td colspan="2">Textbook :

1. Behrouz A. Forouzan, Data Communications & Networking, 5th ed., McGraw-Hill, 2012.

Reference Books:

1. Behrouz A. Forouzan, Computer Networks: A Top-Down Approach, McGraw-Hill, 2012.
2. William Stallings, Data and Computer Communications, 9th ed., Pearson/ Prentice-Hall, 2012.
3. Behrouz A. Forouzan, Data Communications & Networking, 6th ed., McGraw-Hill, 2022.
</td></tr>
</table>

July 2025